Appendix F. The Mathematics of Latency Alerting

Formally, assume that there is a set of user-generated workloads $w \in W$ that are independently run on a service. Additionally, there is some metric $m{:}W \to R$ that measures the performance of the workload on the service, such as latency.

You partition the workloads W into N distinct and disjoint cohorts C_i, where $1 \leq i \leq N$, such that $\cup_{i \leq N} C_i = W$.

Because each workload is (essentially) unique—they may write to different files, compute on different data, be executed on different clusters, etc.—you must cluster workloads in a fuzzy way that captures human expectations of equivalence. Cohorts need to be fine-grained enough to separate workloads with distinct behavior and coarse enough to maintain a reasonable signal-to-noise ratio. A cohort C_i is considered "meaningful" if it meets a minimum cardinality threshold B (e.g., $B \geq 10,000$), and its variance is small relative to its mean:

$$CV(C_i) = \frac{\sqrt{Var(m(C_i))}}{\overline{m}(C_i)} \leq K$$

where:

$$K \leq 0.5$$

The goal is to maximize the number of workloads that fall into these meaningful cohorts; typically, we aim for the union of meaningful cohorts to cover at least 65% of the total workloads, W.

Once cohorts have been defined, compute baseline performance expectations. For simplicity, you can assume that for a given cohort C_i, the performance of its workloads is independent and identically distributed, following an underlying normal distribution:

$$m(w) \sim N(\mu_i, \sigma_i^2) \text{ for } w \in C_i$$

In practice, we use a 30- to 60-day baseline dataset to estimate the mean μ_i and standard deviation σ_i of the measure for each cohort.

This statistical model enables you to compute the likelihood of any particular observation using the standard score, or *z*-score, which measures the number of standard deviations an observation is from the mean:

$$z_w = \frac{m(w)\backslash - \mu_i}{\sigma_i}, \text{ for } w \in C_i$$

For stationary, well-behaved measures, *z*-scores will follow a standard normal ($N(0,1)$) distribution. For a given *z*-score threshold, z_{thr} (e.g., $z_{thr} = 2$), we define our detection metric as the observed proportion of workloads in a test set, W_{test}, with *z*-scores greater than z_{thr}:

$$D(W_{test}, z_{thr}) = \frac{|w \in W_{test} : z_w \; z_{thr}|}{|W_{test}|}$$

For a perfect normal distribution, you would expect 2.28% of *z*-scores to exceed a value of 2. Empirically, we often find this baseline to be slightly higher, around 4%, due to imperfections in the model and natural evolution of the system. A significant increase in this proportion is a strong indicator of a service-wide regression.

This basic statistical technique can be extended. For metrics that are not normally distributed (e.g., latencies are often log-normally distributed), an empirical distribution can be used instead of a parametric one. Furthermore, a hierarchical Bayesian approach can be used to aggregate workloads from smaller, "unmeaningful" cohorts, allowing them to be included in the regression detection model.

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